04:1 Quartiles, Deciles, and Percentiles

- Percentile Location Formula

$L_k = \frac{k}{100}(n+1)$

↳ If L_k is whole number then the percentile is the value at that position

↳ If L_k is falls between two position, the video is averaging those two values

ex:

Position :	1	2	3	4	5	6	7	8	9	10	11	12	13
Value :	9	9	9	9	11	12	12	15	16	16	17	19	20

Median $Q_2 = L_{50} = \frac{50}{100} \times 13 = 7.5 \rightarrow \frac{L_7 + L_8}{2} = \frac{12+15}{2} = \frac{27}{2} = 13.5$

04.3 The Coefficient of Quartile Deviation

Position of $Q_k = \frac{k(n+1)}{4}$

Interquartile Range $IQR = Q_3 - Q_1$

Quartile Deviation (Semi-interquartile range) $QD = \frac{Q_3 - Q_1}{2}$

Coefficient of Quartile Deviation $CQD = \frac{Q_3 - Q_1}{Q_3 + Q_1}$

↳ CQD has no units, so it can compare spread across datasets, like the coefficient of range